Assignment- IX

Group Project: Front End

DUE: 31st October

Total Marks: 10

Submission Instructions

Each group may prepare a single common report. However, every team member must submit the same PDF individually on VTOP.
Include the following details for every team member:

- Name
- Registration Number
- Lab Slot (e.g., L9 + L10)

This submission documents your progress through the **front-end stages** of the ASIC design flow, up to and including Logic Synthesis.

Your report should describe your implementation, present the relevant results, and discuss the findings.

Implementation and Results

Submit a PDF report documenting your implementation and results. Include supporting screenshots, reports, and other relevant evidence.
The report should cover the following, as applicable:

- **RTL Implementation:** Describe the design hierarchy and implementation. Include the relevant RTL modules and explain key design decisions.
- **Functional Verification:** Describe the testbench and verification approach. Present the simulation results and demonstrate that the design meets its functional requirements.
- **Logic Synthesis:** Document the synthesis flow and present the synthesized design results, including relevant area and timing reports.

- **Discussion of Results:** Interpret the results, discuss any issues encountered, and explain how they were resolved. Highlight any important observations from verification and synthesis.

Include sufficient evidence to support your claims. Screenshots should be legible and accompanied by brief explanations; screenshots alone are not sufficient.

Individual Contributions

Clearly describe the contribution of each team member to the front-end implementation. Specify the tasks performed by each member and the work completed.

Note: Contributions may vary across the different stages of the project. A team member who contributes primarily to the back-end implementation will have that contribution considered in the corresponding back-end submission.